

Common Problems

# FB News Feed 🎧

 Scaling ReadsBy Stefan Mai · Published Jul 22, 2024 · 🌟 medium

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## Watch Video Walkthrough

Watch the author walk through the problem step-by-step

 Watch Now

## Understanding the Problem



### What is Facebook's News Feed

Facebook is a social network which pioneered the News Feed, a product which shows recent stories from users in your social graph.

This is a classic system design problem dealing with fan-out and data management. For this problem, let's assume we're handling uni-directional "follow" relationships as opposed to the bi-directional "friend" relationships which were more important for the earliest versions of Facebook.

Let's start by defining our requirements.

## Functional Requirements

### Core Requirements

1. Users should be able to create posts.
2. Users should be able to friend/follow people.
3. Users should be able to view a feed of posts from people they follow, *in reverse chronological order* (newest first).
4. Users should be able to page through their feed.

**Below the line (out of scope):**

- Users should be able to like and comment on posts.
- Posts can be private or have restricted visibility.

For the sake of this problem (and most system design problems for what it's worth), we can assume that users are already authenticated and that we have their user ID stored in the session or JWT.

**Non-Functional Requirements****Core Requirements**

1. The system should be highly available (prioritizing availability over consistency). We'll tolerate up to 1 minute of post staleness (eventual consistency).
2. Posting and viewing the feed should be fast, returning in < 500ms.
3. The system should be able to handle a massive number of users (2B).
4. Users should be able to follow an unlimited number of users, users should be able to be followed by an unlimited number of users.



Having quantities on your non-functional requirements will help you make decisions during your design. A system which is single-digit millisecond fast requires a dramatically different architecture than a "fast" system which can take a second to respond.

Here's how it might look on your whiteboard:

## Functional Requirements

1. Create Posts
2. Follow people
3. View feed
4. Page through feed

## Non-Functional Requirements

- \* Posts visible in  $< 1$  minute (eventually consistent)
- \* Posting and viewing posts  $< 500\text{ms}$
- \* Massive number of users  $2^6$
- \* Unlimited followers/follows

Facebook News Feed Requirements

## The Set Up

### Planning the Approach

The hard part of this design is going to be dealing with the potential of users who are following a massive number of people, or people with lots of followers (a problem known as "fan out"). We'll want to move quickly to satisfy the base requirements so we can dive deep there. For this problem (like many!), following our functional requirements in order provides a natural structure to the interview, so we'll do that.

### Defining the Core Entities

Starting with core entities gives us a set of terms to use through the rest of the interview. This also helps us to understand the data model we'll need to support the functional requirements later.

For the News Feed, the primary entities are easy. We'll make an explicit entity for the link between users:

1. **User**: A user in our system.

2. **Follow:** A uni-directional link between users in our system.
3. **Post:** A post made by a user in our system. Posts can be made by any user, and are shown in the feed of users who follow the poster.

In the actual interview, this can be as simple as a short list like this. Just make sure you talk through the entities with your interviewer to ensure you are on the same page.

User      Follow      Post

Core Entities

## API or System Interface

The API is the primary interface that users will interact with. It's important to define the API early on, as it will guide your high-level design. We can build our API by defining the endpoints necessary for each of our functional requirements.

For our first requirement, we need to create posts:

```
POST /posts
{
  "content": { }
}
// -> 200 OK
{
  "postId": // ...
}
```



We'll leave the content empty to account for rich content or other structured data we might want to include in the post. For authentication, we'll tell our interviewer that authentication tokens are included in the header of the request and avoid going into too much detail there unless requested.

Moving on, we need to be able to follow people. Let's use a simple RESTFUL PUT endpoint for this. The authenticated user (from their JWT) is the one doing the following, and `[id]` is the user they want to follow.

```
PUT /users/[id]/follow
{ }
// -> 200 OK
```



Here the follow action is binary. By using PUT we can ensure it is idempotent so it doesn't fail if the user clicks follow twice. Unfollowing (out of scope) would be accomplished with a DELETE. We don't need a body but we'll include a stub just in case and keep moving.

Our last requirement is to be able to view our feed and page through it.

```
GET /feed?pageSize={size}&cursor={timestamp?}
{
  items: Post[],
  nextCursor: string
}
```



This can be a simple GET request. We've included some parameters to allow the user to page through their feed. Since our requirements are to return posts in reverse chronological order, when users page through their feed they'll be looking at increasingly older posts. We'll use a timestamp as a "cursor" to represent the oldest post they've seen so far, so each page will return N posts older than that timestamp.

We'll avoid diving into the structure of Posts for the moment to give ourselves time for the juicier parts of the interview.



Especially for more senior candidates, it's important to focus your efforts on the "interesting" aspects of the interview. Spending too much time on obvious elements both deprives you of time for the more interesting parts of the interview but also signals to the interviewer that you may not be able to distinguish more complex pieces from trivial ones: a critical skill for senior engineers. "I'll come back if I have time for this" is a great strategy!

Ok, we now have some API's, let's work on building the high-level design behind them.

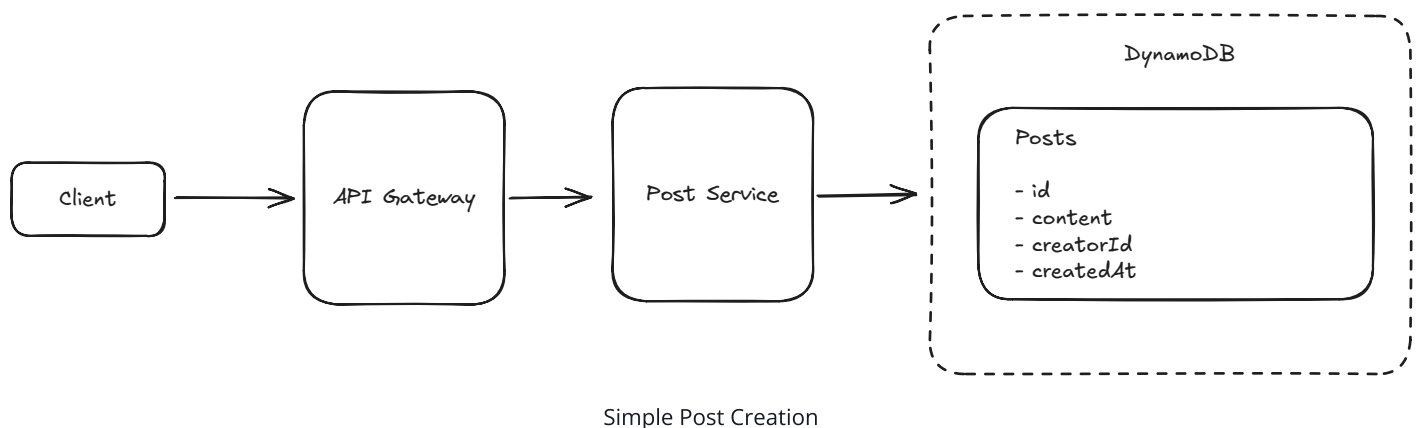
## High-Level Design

## 1. Users should be able to create posts.

In our first requirement, we need to create posts and have them accessible by their ID. We're going to start very basic here and build up complexity as we go.

To start, and since we know this is going to scale, we'll put a horizontally scaled service behind an API gateway/load balancer. We can skip the caching aspect since we'll get to that later. By having the API gateway and load balancer, we can scale up our service with more traffic by adding additional hosts. Since each host is *stateless* as it's only writing to the database (and we're not dealing with reads yet), this is really easy to scale by just adding more hosts!

Users hit our API gateway with a new post request, this gets passed to one of the post service endpoints which creates an insert event that is sent to our database. Easy.



For our database, we can use any key-value store. For this application, let's use DynamoDB for its simplicity and scalability. DynamoDB allows us to provision extremely high throughput provided we spread our load evenly across our partitions.

Great, we can create posts. Onward.

## 2. Users should be able to friend/follow people.

Functionally, following or friending a person is establishing a relationship between two users. This is a many-to-many relationship, and we can model it as a graph. Typically, for a graph you'll use a purpose-built graph database (like Neo4j) or a triple store. However, for our purposes, we can use a simple key-value store and model the graph ourselves.



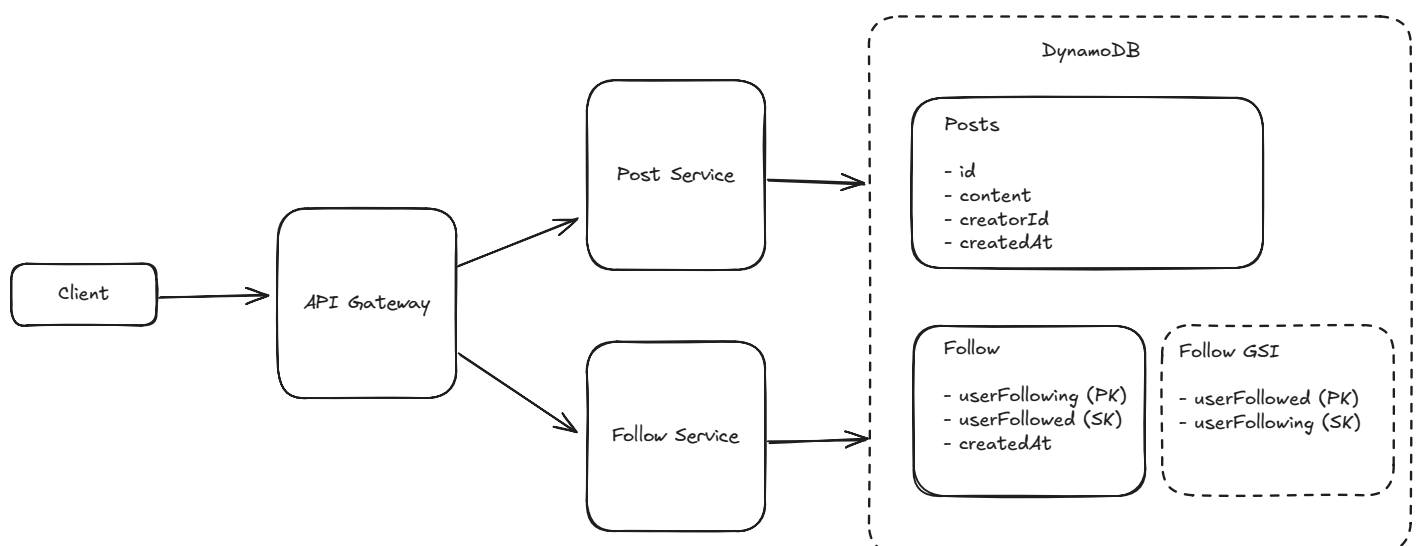
Graph databases can be more useful when you need to do traversals or comprehensions of a graph. This usually requires stepping between different nodes, like capturing the friends of your friends or creating an embedding for a recommendation system. We only have simple requirements in this problem so we'll keep our system simple and save ourselves from scarier questions like "how do you scale Neo4j?" We don't need a full-fledged graph database for this problem.

To do this, we'll have a simple **Follow** table using the entire relation with **userFollowing** as the partition key and **userFollowed** as the sort key. We can also create a Global Secondary Index (GSI) with the reverse relationship (e.g. partition key of **userFollowed** and sort key of **userFollowing**) to allow us to look up all the followers of a given user.

This allows us to query for the important pieces:

- If we want to check if the user is following another user, we query with both the partition key and the sort key (e.g. **userFollowing:userFollowed**). This is a simple lookup!
- If we want to get all the users a given user is following, we query with the partition key (e.g. **userFollowing**). This is a range query.
- If we want to get all the users who are following a given user, we query the GSI with its partition key (e.g. **userFollowed**). This is a range query.

We can put this data in another DynamoDB table for simplicity. In practice, AWS recommends single-table design for DynamoDB, but using separate tables here keeps our discussion clearer and is totally fine for an interview.



We're off to a good start. Let's keep going.

### 3. Users should be able to view a feed of posts from people they follow.

The challenge of viewing a feed of posts can be broken down into steps, which we can do with our present design with only a few changes:

1. First, we get all of the users who a given user is following.
2. Next, we get all of the posts from those users.
3. Finally, we sort *all* those posts by time and return them to the user.

We'll start by adding a feed service to do this work. It's going to be read-heavy and have very different query patterns than the Post and Follow services, so separating it out makes sense.

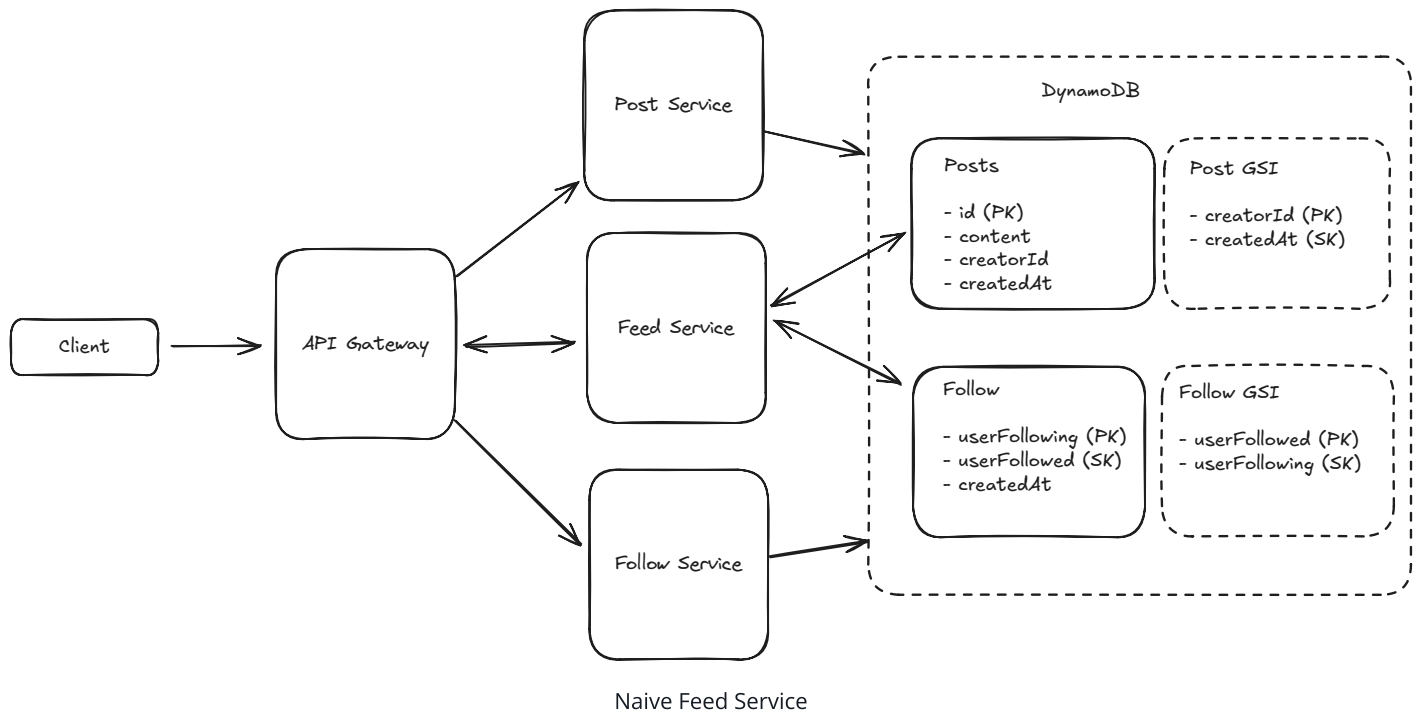
#### Pattern: Scaling Reads

News feed generation is quintessentially read-intensive - users check their feeds constantly but post rarely. This makes **scaling reads** essential through pre-computing feeds for active users, caching recent posts from followed users, and smart pagination. The key is that users typically only read the first few items, so aggressive caching of recent content delivers massive performance gains.

[Learn This Pattern](#)

Finding all the posts from a given user is tricky with the existing Posts table: while we have an index on our follow table to quickly look up all the users a user is following, we don't have an index on our post table to quickly look up all the posts from a set of users. We can solve this by adding a GSI to the `Post` table with the `creatorID` as the partition key and the post's creation timestamp `createdAt` as the sort key. This will allow us to quickly pull all the posts from a set of users in chronological order.





That seems to work. For a given user, we first request all of the users they follow from the **Follow** table. Then we request all the posts from those N users from the **Post** table via its GSI. Then we sort all of those posts by timestamp and return the results to the user!

Now alarm bells may be ringing for you and that's good. Several flags emerge here and your interviewer is going to expect you to see them quickly, so it's good to note them verbally:

1. Our user may be following lots of users.
2. Each of those users may have lots of posts.
3. The total set of posts may be very large because of (1) or (2).

We'll need to solve those before we wrap things up! But before we dive into these complexities let's polish off our functional requirements.



In a real interview, I'd explain this to the interviewer exactly like that "I know this is not going to scale but I'm going to start with a naive solution and then solve the scaling problems separately".

A very common failure mode for candidates we work with is that they get lost in the weeds of a particular scaling problem before having a (mostly) complete design. By covering the breadth of our requirements before we go into depth we can avoid this situation - as long as we keep moving quickly!

## 4. Users should be able to page through their feed.

We want to be able support an "infinite scroll" -like experience for our users. To do this, we need to know what they've already seen in their feed and be able to quickly pull the next set of posts. Fortunately, "what they've already seen in their feed" can be described fairly concisely: a timestamp of the oldest post they've looked at. Since users are viewing posts from youngest to oldest, that one timestamp tells us where they've stopped and provides a great cursor for the next set of posts.

Since we already have a GSI on the post table which sorts posts by creation timestamp, we can use that directly to return only posts older than the cursor. This looks just like the previous functional requirement with one addition:

1. First, we get all of the users who a given user is following.
2. Next, we get all of the posts from those users **that are older than the input timestamp**.
3. Finally, we sort *all* those posts by time and return them to the user.

Great! We have a working system that allows a small number of users to follow a small number of users with a small number of posts and view their feeds!

Now let's circle back so some of the issues we flagged earlier in our deep dives.

## Potential Deep Dives

### 1) How do we handle users who are following a large number of users?

First, if a user is following a large number of users, the queries to the `Follow` table will take a while to return. Once we get each of the people they're following, we'll be making a large number of queries to the `Post` table to build their feed. This is called **fan-out on read** — a single read request fans out to create many more requests. Especially when latency is a concern, fan-out on read can be a real problem, so it makes for a good discussion point in system design interviews.

It's not uncommon to generate 10s to 100s of requests to satisfy an incoming request, but it is rarer to have to generate 1000's of requests, especially for a service that needs to serve users with low latency.

So what can we do instead? Your instinct here should be to think about ways that we can compute the feed results on *write* or post creation rather than at the time we want to read their feed. This is called **fan-out on write** — instead of assembling the feed when a user asks for it, we precompute feeds when posts are created. And in fact this is a good line of thinking!



While this particular question begs the question of how to handle a large number of follows, a sensible question for your interviewer might be "can we adjust the product in these instances?" Facebook, for example, sets the max number of friends as 5,000. Setting a max number of follows, or having a slightly different experience for these users is a very common approach for production systems like this. Is your user with 100k follows really going to notice if their posts are appearing a couple minutes late?

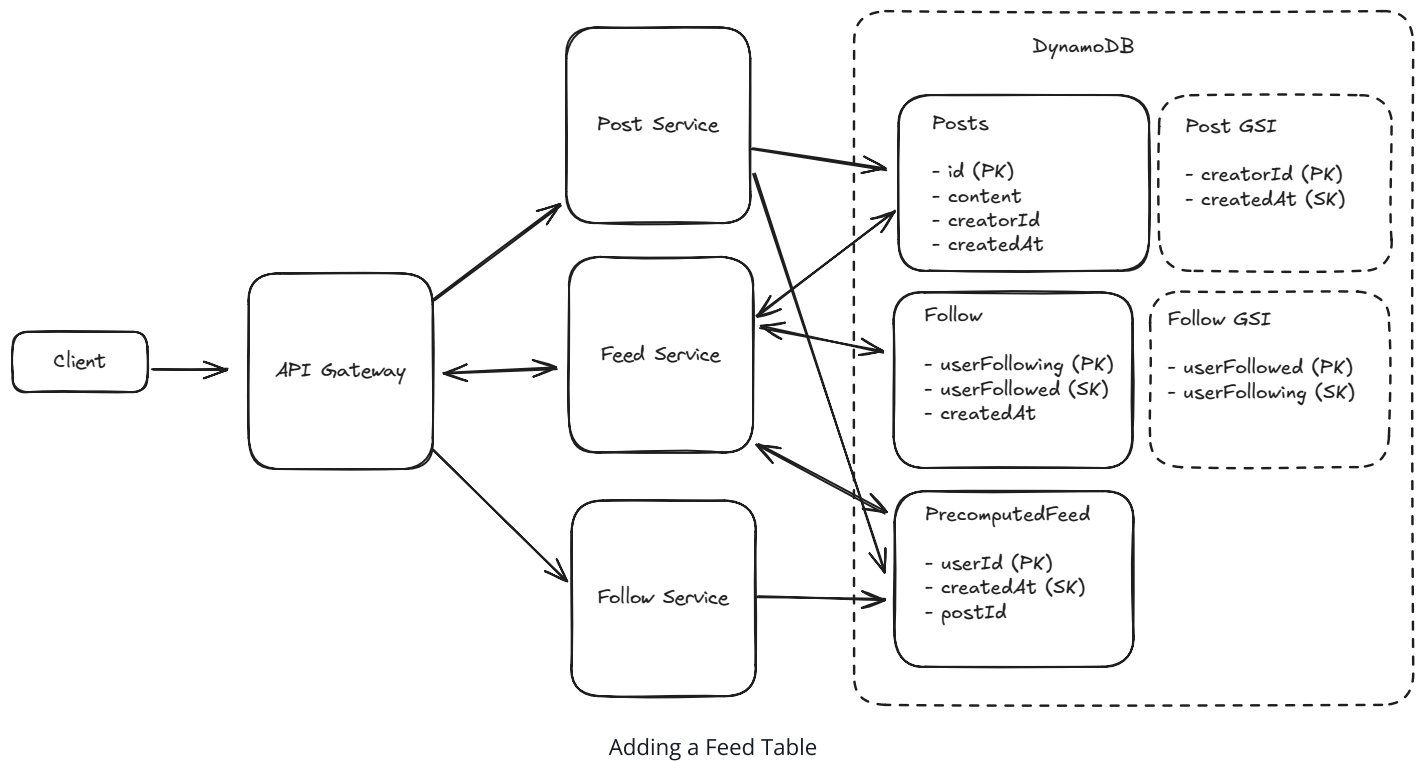
For users following a large number of users we can keep a **PrecomputedFeed** table. Instead of querying the **Follow** and **Post** tables, we'll be able to pull from this precomputed table. Then, when a new post is created, we'll simply add to the relevant feeds.

Each entry in the **PrecomputedFeed** table is keyed by **userId** and its value is a list of post IDs in reverse chronological order, limited to a small number of posts (say 200 or so). We want each entry to be compact so we can minimize the amount of space we require. Since we only ever access this table by user ID, we don't need to deal with any secondary indexes.



People frequently wonder "what about if users page back 100 pages into {their feed, search results, etc}?" While this is a fair question, most systems will simply just not support this (try to go a few dozen pages deep into Google) because real users aren't doing this.

At the end of the day, this fixed 200 number here could be increased or decreased by how much it impacts those tail users and the relative tradeoff in cost. We can always fall back to our naive solution of querying the base **Follow** and **Post** tables if we need to get more content for users who have reached the end of their feed.



Let's quickly gut-check the storage required of this solution before we move on. Let's assume each postID is 10 bytes, to store 200 posts per user we'd need 2KB per user. If we have 2B users, we need 4TB of storage for all these feeds. This is quite reasonable for a modern system.



One useful rule-of-thumb is to ask how much a given user, tenant, or item is going to cost in dollars. 2kb of data is a fraction of a cent per month and Facebook earns something like \$100 per year for US-based users. So this is pretty reasonable.

While this dramatically improved read performance, it creates a new problem: when a user with a lot of followers creates a post, we need to write to millions of feeds efficiently. So let's dive into that next.

## 2) How do we handle users with a large number of followers?

When a user has a large number of followers we have a similar fanout problem when we create a post: we need to *write* to millions of **Feed** records!

Because we chose to allow some inconsistency in our non-functional requirements, we have a short window of time (< 1 minute) to perform these writes.

**Bad Solution: Blast the Requests**



**Good Solution: Async Workers**



**Great Solution: Async Workers with Hybrid Feeds**



### 3) How can we handle uneven reads of Posts?

To this point, our feed service has been reading directly from the Post table whenever a user requests to get their feed. For the vast majority of posts, they will be read for a few days and never read again. For some posts (especially those from accounts with lots of followers), the number of reads the post experiences in the first few hours will be *massive*.

DynamoDB, like many key-value stores, can scale to very high throughput *provided certain conditions are met*. One of the more important conditions is there being even load across the keyspace. Under the covers DynamoDB has physical machines with real limitations like any other database. If Post1 gets 500 requests per second and Post2 through Post 1000 get 0 requests per second, this is not even load!

Fortunately for us, Posts are far more likely to be created than they are to be edited. But how do we solve the issue of hot keys in our Post Table?

**Good Solution: Post Cache with Large Keyspace**



**Great Solution: Redundant Post Cache**



## What is Expected at Each Level?

Ok, that was a lot. You may be thinking, "how much of that is actually required from me in an interview?" Let's break it down.

### Mid-level

**Breadth vs. Depth:** A mid-level candidate will be mostly focused on breadth (80% vs 20%). You should be able to craft a high-level design that meets the functional requirements you've defined, but many of the components will be abstractions with which you only have surface-level familiarity.

**Probing the Basics:** Your interviewer will spend some time probing the basics to confirm that you know what each component in your system does. For example, if you add an API Gateway, expect that they may ask you what it does and how it works (at a high level). In short, the interviewer is not taking anything for granted with respect to your knowledge.

**Mixture of Driving and Taking the Backseat:** You should drive the early stages of the interview in particular, but the interviewer doesn't expect that you are able to proactively recognize problems in your design with high precision. Because of this, it's reasonable that they will take over and drive the later stages of the interview while probing your design.

**The Bar for News Feed:** For this question, an E4 candidate will have clearly defined the API endpoints and data model, and landed on a high-level design that is functional and meets the requirements. While they may have some of the "Good" solutions, they would not be expected to cover all the possible scaling edge cases in the Deep Dives.

## Senior

**Depth of Expertise:** As a senior candidate, expectations shift towards more in-depth knowledge — about 60% breadth and 40% depth. This means you should be able to go into technical details in areas where you have hands-on experience. It's crucial that you demonstrate a deep understanding of key concepts and technologies relevant to the task at hand.

**Advanced System Design:** You should be familiar with advanced system design principles. For example, knowing approaches for handling fan-out is essential. You'll also be expected to iteratively diagnose performance bottlenecks and suggest improvements. Your ability to navigate these advanced topics with confidence and clarity is key.

**Articulating Architectural Decisions:** You should be able to clearly articulate the pros and cons of different architectural choices, especially how they impact scalability, performance, and maintainability. You justify your decisions and explain the trade-offs involved in your design choices.

**Problem-Solving and Proactivity:** You should demonstrate strong problem-solving skills and a proactive approach. This includes anticipating potential challenges in your designs and suggesting improvements. You need to be adept at identifying and addressing bottlenecks, optimizing performance, and ensuring system reliability.

**The Bar for News Feed:** For this question, E5 candidates are expected to speed through the initial high level design so you can spend time discussing, in detail, at least 2 of the deep dives. They should also be able to discuss the pros and cons of different architectural choices, especially how they impact scalability, performance, and maintainability. They would be expected to proactively surface some of the potential issues around fanout and performance bottlenecks.

## Staff+

**Emphasis on Depth:** As a staff+ candidate, the expectation is a deep dive into the nuances of system design — I'm looking for about 40% breadth and 60% depth in your understanding. This level is all about demonstrating that, while you may not have solved this particular problem before, you have solved enough problems in the real world to be able to confidently design a solution backed by your experience.

You should know which technologies to use, not just in theory but in practice, and be able to draw from your past experiences to explain how they'd be applied to solve specific problems effectively. The interviewer knows you know the small stuff (REST API, data normalization, etc) so you can breeze through that at a high level so you have time to get into what is interesting.

**High Degree of Proactivity:** At this level, an exceptional degree of proactivity is expected. You should be able to identify and solve issues independently, demonstrating a strong ability to recognize and address the core challenges in system design. This involves not just responding to problems as they arise but anticipating them and implementing preemptive solutions. Your interviewer should intervene only to focus, not to steer.

**Practical Application of Technology:** You should be well-versed in the practical application of various technologies. Your experience should guide the conversation, showing a clear understanding of how different tools and systems can be configured in real-world scenarios to meet specific requirements.



**Complex Problem-Solving and Decision-Making:** Your problem-solving skills should be top-notch. This means not only being able to tackle complex technical challenges but also making informed decisions that consider various factors such as scalability, performance, reliability, and maintenance.

**Advanced System Design and Scalability:** Your approach to system design should be advanced, focusing on scalability and reliability, especially under high load conditions. This includes a thorough understanding of distributed systems, load balancing, caching strategies, and other advanced concepts necessary for building robust, scalable systems.

**The Bar for News Feed:** For a staff+ candidate, expectations are high regarding depth and quality of solutions, particularly for the complex scenarios discussed earlier. A staff candidate will likely cover all of the deep dives (and/or some that we haven't enumerated). They would be expected to surface potential issues in the system and talk about performance tuning for this problem.

## Changelog

- 2024/07/22: Initial version
- 2025/06/03: Made updates to address common questions in comments
- 2026/02/17: Fixed reverse chronological order terminology, corrected storage calculation, removed erroneous "Redis again" reference, clarified follow API endpoint, added fan-out on read/write terminology, improved cache strategy descriptions, noted DynamoDB single-table design best practice, and various grammar fixes.

### Test Your Knowledge

Take a quick 15 question quiz to test what you've learned and identify gaps.

 Start Quiz